BlueROV2 Localization Methods Research

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Contents

I. Abstract

This document investigates possible localization and obstacle avoidance methods for the BlueROV2. The current BlueROV2 system does not have a reliable method for tracking and mapping its position, which limits the ability to visualize path smoothness, evaluate convergence toward a sound source, and support obstacle avoidance. This document is structured in the following order: the project objective and relevant underwater sensing challenges are addressed, all available sonar types are described and compared, then several possible localization methods are discussed and finally the recommended next step is outlined.

II. Objective

The current BlueROV2 system does not have a method for tracking and mapping its position during operation. Implementing a tracking and mapping system is necessary to visualize the smoothness of the BlueROV2's path and its convergence toward the sound source in ONR reports. Additionally, this tracking system must be able to support future obstacle avoidance capabilities.

2.1 Requirements

There are two requirement for the system that must be met:

1. **System Versatility**
   The tracking system must be usable in both pool and ocean environments.

2. **Sensor Coverage**
   The sensors must be capable of covering the full area of travel of the BlueROV2 in order to detect and avoid obstacles.

2.2 Standing Challenges

In developing this system, three primary challenges must be addressed:

1. **Drift**
   This is a primary issue with sensors such as inertial measurement units (IMUs), which can accumulate error over time and eventually output inaccurate measurements.

2. **Water properties**
   Sensors such as LiDAR and cameras may produce inaccurate readings underwater. At greater depths, laser-based measurements can become unreliable because light is scattered, reflected, and attenuated by the water. Cameras may also experience distorted depth perception due to refraction, which can lead to errors in image processing. In addition, low-light conditions at larger depths may make visual sensing difficult or impractical.

3. **Servo noise**
   The servos generate noise underwater. If sonar or another acoustic sensor is used, interference from servo noise may affect measurement accuracy. Therefore, data filtering or noise-reduction techniques may need to be implemented.

III. Sonar Types

Before discussing plausible solutions and research methodologies, this section introduces the main types of sonar. This background will make the following sections easier to understand and evaluate.

3.1 What Are Sonars?

Simply put, sonars are devices that transmit sound waves through water. These sound waves travel outward, reflect off nearby surfaces or objects, and return to the sonar device. The device then uses the return signal to estimate the distance between itself and the object. This allows underwater geometry, obstacles, and terrain features to be detected and used for navigation, mapping, and tracking.

There are two primary types of sonar commonly available: **echo sounders** and **imaging sonars**. In general, echo sounders collect distance or depth data points, while imaging sonars use many acoustic returns to generate image-like representations of the underwater environment.

3.2 Echo Sounders

Echo sounders are sonar devices that transmit sound waves and collect distance or depth measurements. These measurements can then be processed for several purposes, including tracking and mapping. Echo sounders are often discussed in the context of bathymetry, which refers to measuring underwater depth or terrain.

1. **Single-Beam Echo Sounder**

   Single-beam echo sounders transmit one narrow acoustic beam and provide a single depth measurement point at a time.

   Pros:

   (a) **Versatility:** Single-beam echo sounders are often available in both high-frequency and low-frequency configurations.

   (b) **Low Cost:** This is typically the least expensive type of echo sounder, with options starting at approximately $890.

   (c) **Compact Size:** Single-beam echo sounders are generally the most compact type of echo sounder.

   (d) **Ease of Use:** Since the sonar only collects a single data point at a time, the data processing requirements are relatively simple.

   Cons:

   (a) **Single Data Point:** Because only one data point is gathered and the sensor does not scan across a wider field of view, only a very small area is covered. This limits its usefulness for obstacle avoidance.

2. **Multibeam Echo Sounder**

   Multibeam echo sounders transmit multiple acoustic beams simultaneously. This allows a wider area to be measured at once compared to a single-beam echo sounder.

   Pros:

   (a) **Wide Tracking Range:** A typical multibeam echo sounder can track a wide angular region, often up to approximately 80°, which improves obstacle detection and mapping capabilities.

   (b) **Large Dataset:** Because several beams are transmitted and measured simultaneously, the sonar gathers a significantly larger dataset. This can provide more meaningful information for obstacle avoidance and mapping.

   Cons:

   (a) **Price:** A multibeam echo sounder starts at approximately $4,990, which is significantly more expensive than a single-beam echo sounder.

(b) **Size:** A multibeam echo sounder is generally much larger than a single-beam echo sounder, which may make mounting on the BlueROV2 more difficult.

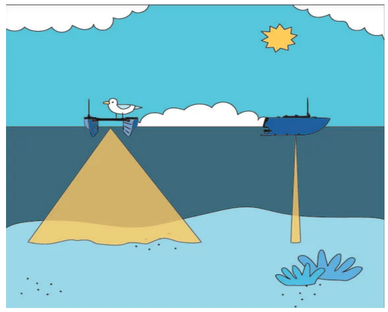

Figure 1: Multibeam echo sounder, left, compared to single-beam echo sounder, right, from Blue Robotics.

3.3 Imaging Sonars

Imaging sonars are sonar devices that transmit sound waves and use the returning echoes to form image-like representations of the underwater environment. Unlike echo sounders, which primarily collect distance or depth measurements, imaging sonars gather acoustic data across a wider field of view. These measurements can then be processed to identify objects, obstacles, terrain features, and the relative position of the BlueROV2.

1. **Scanning Imaging Sonar**

   A scanning imaging sonar uses a single acoustic beam to produce an image of the surrounding underwater terrain. This beam rotates through a selected scan angle, up to 360°, allowing a map of the surfaces surrounding the BlueROV2 to be constructed.

   Pros:

   (a) **Holistic View:** Because this sonar can provide a view of the surrounding environment, obstacles can be tracked and avoided more easily.

   (b) **Price:** This sonar is relatively inexpensive compared to other imaging sonar options, with prices starting at approximately $2,750.

   (c) **Image Quality:** This sonar offers high-resolution images, which can make localization and obstacle avoidance easier.

   Cons:

   (a) **Scan Speed:** This sonar can take between 5 and 20 seconds to perform a scan, depending on the selected scan angle. This long scan time limits the speed at which the BlueROV2 can move while still collecting useful data.

2. **Multibeam Imaging Sonar**

   A multibeam imaging sonar has a similar working principle to a multibeam echo sounder. These are forward-looking sonars that simultaneously transmit several acoustic beams to scan a wide angle, often around 90°, and image the terrain directly ahead of the BlueROV2.

   Pros:

   (a) **Wide Horizontal Coverage:** With an approximately 90° scan angle, obstacle avoidance becomes significantly easier.

   (b) **Fast Update Rate:** Because the beams are transmitted simultaneously, the data update rate is fast, allowing the BlueROV2 to travel at higher speeds.

   (c) **High Resolution:** Like scanning imaging sonar, multibeam imaging sonar can produce high-resolution images, which supports obstacle avoidance and tracking.

Cons:

(a) **Price:** Multibeam imaging sonars can start at approximately \$9,000, making them a very expensive option.

(b) **Vertical Coverage:** These sonars are limited in terms of vertical coverage, often reaching only about 20°. Therefore, strategic placement of the sonar on the BlueROV2 is crucial.

3. **Side-Scan Imaging Sonar**

Side-scan imaging sonars are mounted to the side of the BlueROV2 and scan the surrounding environment as the vehicle moves forward. These sonars use a fan-shaped beam, often with a vertical beam height of approximately 50°, to image large areas along the side of the vehicle.

Pros:

(a) **Scan Range:** These sonars can cover a very large area, which can be useful for mapping and obstacle detection.

(b) **Price:** These sonars start at approximately \$2,950, which is relatively inexpensive if only one unit is used.

(c) **High Resolution:** Like scanning imaging sonar, side-scan sonar can produce high-resolution images, which can support obstacle detection and tracking.

Cons:

(a) **Limited Effectiveness in Shallow Water:** In shallow water, these sonars can be limited by the amount and quality of data collected.

(b) **Unintuitive Imagery:** Because the BlueROV2 may move along a non-smooth path due to its thruster configuration, the resulting side-scan images may be distorted or difficult to process for obstacle avoidance and localization.

Table 1: Sonar hardware compared with the type of data each sonar produces.

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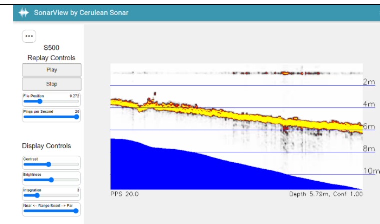
<tr><td>Single-Beam Echo Sounder</td><td></td><td>
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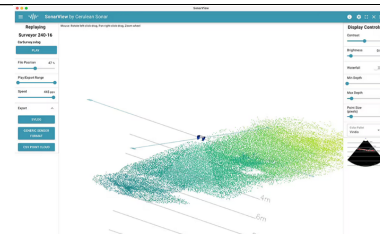
<tr><td>Multibeam Echo Sounder</td><td></td><td>
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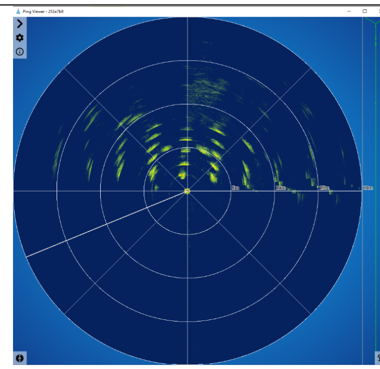
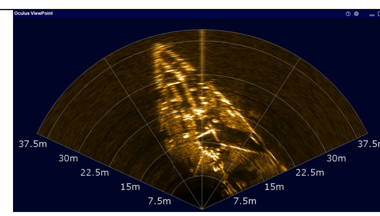
<tr><td>Scanning Imaging Sonar</td><td></td><td></td></tr>
<tr><td>Multibeam Imaging Sonar</td><td></td><td>
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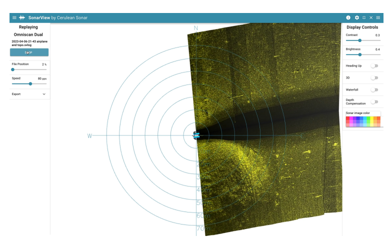
<tr><td>Side-Scan Imaging Sonar</td><td></td><td>
</td></tr>
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IV. Possible Solutions

The main goal of this project is to develop a tracking and mapping system for the BlueROV2 that can operate in both pool and ocean environments. This system should allow the BlueROV2's path to be visualized, make it possible to evaluate convergence toward a sound source, and eventually support obstacle avoidance. Since underwater localization is difficult due to sensor drift, poor visibility, acoustic interference, and changing water properties, several possible solutions must be considered.

4.1 Research-Backed Methods

1. **IMU, Imaging Sonar, and Camera Integration**
   Reference: https://arxiv.org/html/2503.01434v1

   This method would use the existing IMU and camera on the BlueROV2 along with an imaging sonar. The IMU would provide motion estimates, while the camera and imaging sonar would collect environmental data for feature recognition and matching. By comparing features across multiple frames or scans, the system could correct some of the accumulated localization error caused by IMU drift.

   Pros:

   (a) **Uses Existing Sensors:** Since the BlueROV2 already has an IMU and camera, only the imaging sonar would need to be added.

   (b) **Feature-Based Correction:** Camera and sonar data can be used to recognize environmental features, which may help correct IMU drift.

   (c) **Obstacle Avoidance Potential:** The imaging sonar can provide information about surrounding objects, making this setup useful for future obstacle avoidance.

   (d) **Moderate Cost:** Imaging sonars start at approximately $2,500, making this option less expensive than systems requiring several high-cost sensors.

   Cons:

   (a) **IMU Drift:** The IMU will still accumulate error over time, so the system depends heavily on how well the camera and sonar data can correct this drift.

   (b) **Camera Limitations:** The camera may perform poorly in low-light, murky, or deep-water environments.

   (c) **Processing Complexity:** Feature recognition and matching require significant image and sonar processing.

   (d) **Environmental Dependence:** This method may perform poorly in uniform or featureless environments.

2. **IMU, Imaging Sonar, Camera, and Pressure Sensor Integration**
   Reference: https://arxiv.org/pdf/1810.03200

   This method expands on the previous approach by adding the BlueROV2's pressure sensor. The IMU would provide orientation and motion estimates, the imaging sonar and camera would support feature detection, and the pressure sensor would provide depth measurements. This method could also use image preprocessing techniques, such as Contrast Limited Adaptive Histogram Equalization, to improve image quality and increase the reliability of feature detection.

   Pros:

   (a) **Improved Depth Estimation:** The pressure sensor can provide a direct measurement of depth, improving the vertical position estimate of the BlueROV2.

   (b) **Better Error Correction:** Combining IMU, sonar, camera, and pressure data gives the system more information to correct localization errors.

   (c) **Improved Image Processing:** Image preprocessing methods can improve feature detection, especially in underwater images with poor contrast.

(d) **Uses Existing BlueROV2 Sensors:** The IMU, camera, and pressure sensor are already available on the BlueROV2, so the main additional hardware would be the imaging sonar.

Cons:

(a) **Higher System Complexity:** Combining data from multiple sensors requires sensor fusion, calibration, and synchronization.

(b) **Camera Reliability:** Even with preprocessing, the camera may still struggle in dark, murky, or low-visibility environments.

(c) **Sonar Cost:** The system still requires an imaging sonar, which adds at least approximately $2,500 to the project cost.

(d) **Implementation Difficulty:** Developing a reliable localization algorithm using several sensors may require significant testing and tuning.

3. **Sonar-Based SLAM**
Reference: https://github.com/jake3991/sonar-SLAM

Sonar-based SLAM uses dead reckoning to estimate the initial position of the vehicle and then corrects drift using sonar scan matching. One documented approach uses Iterative Closest Point, or ICP, to match sonar scans and reduce localization error. This method would require an IMU and DVL for dead reckoning, along with an imaging sonar for mapping and scan matching.

Pros:

(a) **Mapping and Localization:** This method can estimate the vehicle's position while also building a map of the surrounding environment.

(b) **Drift Correction:** ICP can help reduce drift by aligning sonar scans over time.

(c) **Useful for Obstacle Avoidance:** Since the method builds a map, it could support obstacle detection and path planning.

(d) **Research Support:** Existing documented work and open-source implementations are available, which could help guide development.

Cons:

(a) **Requires a DVL:** The method uses a DVL for dead reckoning, which increases system cost. One lower-cost DVL option is approximately $3,000.

(b) **2D Limitation:** Documented work is primarily limited to a 2D plane, meaning pitch and roll cannot be fully included.

(c) **Computational Complexity:** ICP and sonar scan matching can be computationally demanding.

(d) **Implementation Difficulty:** Adapting this method to the BlueROV2 and validating it in both pool and ocean environments may be challenging.

4. **AUV Positioning Using Object and Place Recognition**
Reference: https://arxiv.org/pdf/2405.01971v1

This method uses a map of the seabed and 3D CAD models of known underwater assets to localize the vehicle. A multibeam forward-looking sonar would collect environmental data, and object or place recognition models, such as YOLO, would identify known features or assets. The AUV could then estimate its position based on the recognized objects and their known locations on the map.

Pros:

(a) **Feature-Based Localization:** Recognizing known objects or places can provide strong localization references.

(b) **Useful in Structured Environments:** This method could work well in areas with known assets, such as underwater infrastructure or pre-mapped test sites.

(c) **Advanced Recognition Capability:** Object detection models can identify specific objects rather than only measuring distance.

(d) **Reduced Dependence on Dead Reckoning:** Recognized objects can act as position references and help correct accumulated drift.

Cons:

(a) **Requires Prior Maps:** The method requires a map of the seabed before localization can be performed.

(b) **Requires CAD Models:** The system also requires 3D CAD models of the underwater assets it is expected to recognize.

(c) **Limited Generalization:** This method may not work well in unknown or unstructured environments.

(d) **High Sensor Cost:** A multibeam forward-looking sonar is required, which can be expensive.

4.2 Alternate Options

In addition to the research-based methods, there are several alternate options that may be useful depending on the testing environment, budget, and required localization accuracy. These options are generally simpler than full SLAM or multi-sensor fusion methods, but they may also provide less complete tracking or mapping capability.

1. **ROV Locator System**

   An ROV locator system uses a transmitter and receiver to track the real-time position of the BlueROV2 relative to a fixed receiver. In this setup, the receiver would be placed at a known static location, and the transmitter would be mounted on the ROV. This option is relatively simple and provides direct position tracking, but it does not create a map of the surrounding environment or directly supports obstacle avoidance.

2. **Modified Fishing Sonar**

   Another possible option is modifying an existing fishing sonar for BlueROV2 localization and mapping. Fishing sonars are designed to detect underwater objects, fish, and terrain, so they may provide useful acoustic data at a lower cost than research-grade sonar systems. However, this option may be difficult because fishing sonar data is often processed automatically through proprietary software, making raw data access and software integration difficult.

3. **Pool Localization Using Single-Beam Echo Sounders**

   For pool testing, a simpler localization method could be developed using single-beam echo sounders. If the geometry of the pool is known, such as through a CAD model, the echo sounder measurements could be compared against the expected distances to the pool walls or floor. These measurements could provide an approximate x-y position, while the pressure sensor could provide the z position. This option is low-cost and useful for controlled pool testing, but it would likely not work well to ocean environments and would not fully support obstacle avoidance.

V. Recommended Next Steps

Based on the localization and obstacle avoidance requirements, the recommended approach is to separate the system into two parts: one system for tracking the BlueROV2's position and another system for detecting obstacles. For localization, the best next step is to use the ROV Locator System. This option provides direct real-time tracking relative to a fixed receiver, which makes it practical for both testing and reporting. It would allow the BlueROV2's path smoothness and convergence toward the sound source to be visualized without requiring a full SLAM system which is both expensive and difficult to implement.

For obstacle avoidance, a sonar-based solution is recommended because cameras and LiDAR are less reliable underwater. To properly avoid obstacles, the BlueROV2 should have close to a 360° view of its surroundings. One possible solution is to mount two Ping360 scanning sonars, one at the front and one at the back of the BlueROV2. Each sonar could scan 180°, which would provide full surrounding coverage while taking approximately half the time required for one sonar to complete a full 360° scan.

Another lower-cost option is to mount single-beam echo sounders on servos, similar to the hydrophone mounting approach. By rotating the echo sounders, the system could generate a 360° point cloud around the BlueROV2. This would provide useful obstacle information, but the mechanical design, scan speed, and point cloud density would need to be tested.

Side-scan sonars could also be considered because they provide wide coverage. However, they are less ideal for this application because the data is image-based rather than point-cloud-based, making obstacle avoidance and localization processing more difficult. The most ideal sonar solution would be a multibeam echo sounder mounted in a forward-looking direction, since it provides both a point cloud and wide coverage. However, this option would likely be more expensive and may require careful mounting to ensure useful coverage around the BlueROV2.

It is important to note that the BlueROV2 has an internal measurement unit (IMU).

VI. (Pulled from BlueROV2 Assembly Instructions) Potential Areas for Improvement

6.1 Wiring & Packaging

- The current electronics packaging is tightly constrained, which prevents additional components, such as sonar for tracking, from being integrated. Potential improvements include:
    1. Designing a custom PCB for power distribution to reduce jumper-wire volume and improve wire organization inside the clear housing.
    2. Replacing the Arduino UNO with an PWM expansion/Auxiliary (direct connection to Raspberry Pi instead) board to reduce the electronics footprint.
    3. Replacing the current hydrophones with models that have shorter cables, ideally no longer than half a meter and that have integrated preamplifier.
    4. Redesigning the internal 3D-printed electronics frame to better conform to the clear housing and increase usable space.
    5. Improve o-ring connection between hydrophone hose and hydrophone end.
    6. Look for method to mount an additional clear housing for additional sensor implementation.

VII. Resources

1. **Single-Beam Echo Sounder:**
   https://ceruleansonar.com/product/sounder-s500/
2. **Multibeam Echo Sounder:**
   https://ceruleansonar.com/product/surveyor-240-16/
3. **Scanning Imaging Sonar:**
   https://bluerobotics.com/store/sonars/imaging-sonars/ping360-sonar-r1-rp/
4. **Multibeam Imaging Sonar:**
   https://www.blueprintsubsea.com/oculus/
5. **Side-Scan Imaging Sonar:**
   https://ceruleansonar.com/product/omniscan-450ss-450ss-blueboat/
6. **DVL:**
   https://ceruleansonar.com/product/tracker-650-cmp/
7. **ROV Locator System:**
   https://ceruleansonar.com/product/rov-locator-mark-ii/
8. **IMU, Imaging Sonar, and Camera Integration Paper:**
   https://arxiv.org/html/2503.01434v1
9. **IMU, Imaging Sonar, Camera, and Pressure Sensor Integration Paper:**
   https://arxiv.org/pdf/1810.03200
10. **Sonar-Based SLAM Repository:**
    https://github.com/jake3991/sonar-SLAM
11. **AUV Positioning Using Object and Place Recognition Paper:**
    https://arxiv.org/pdf/2405.01971v1